

Panel Data Models: RE and FE

ECMT 676 (Econometric II) Lecture Notes

TAMU

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Roadmap

- Random effects estimation
- Fixed effects estimation
- First-difference estimation

- We already covered panel data models where the error term had no particular structure.
- But we assumed either contemporaneous exogeneity (pooled OLS) or strict exogeneity (feasible GLS).
- Now we explicitly add a time constant, *unobserved effect* to the model. Often called *unobserved heterogeneity*.
- Start with the balanced panel case, and assume random sampling across i (the cross section dimension), with fixed time periods T . So $\{(\mathbf{x}_{it}, y_{it}) : t = 1, \dots, T, c_i\}$ where c_i is the unobserved effect drawn along with the observed data.
- The unbalanced case is trickier because we must know why we are missing some time periods for some units. We consider this much later under missing data/sample selection issues.

- For a random draw i from the population, the basic model is

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}, \quad t = 1, \dots, T,$$

where $\{u_{it} : t = 1, \dots, T\}$ are the *idiosyncratic errors*. The *composite error* at time t is

$$v_{it} = c_i + u_{it}$$

- Because of c_i , the sequence $\{v_{it} : t = 1, \dots, T\}$ is almost certainly serially correlated, and definitely is if $\{u_{it}\}$ is serially uncorrelated.

- Useful to write a population version of the model in conditional expectation form:

$$E(y_t|\mathbf{x}_t, c) = \mathbf{x}_t\boldsymbol{\beta} + c, \quad t = 1, \dots, T.$$

Therefore,

$$\beta_j = \frac{\partial E(y_t|\mathbf{x}_t, c)}{\partial x_{tj}},$$

so that β_j is the partial effect of x_{tj} on $E(y_t|\mathbf{x}_t, c)$, so that we are “holding c fixed.”

- Hope is that we can allow c to be correlated with $\mathbf{x}_t$.

- With a single cross section, there is nothing we can do unless we can find good observable proxies for c or IVs for the endogenous elements of $\mathbf{x}_t$. But with two or more periods we have more options.
- We can write the population model as

$$\begin{aligned}y_t &= \mathbf{x}_t\boldsymbol{\beta} + c + u_t \\ E(u_t|\mathbf{x}_t, c) &= 0\end{aligned}$$

Suppose we have $T = 2$ time periods:

$$\begin{aligned}y_1 &= \mathbf{x}_1\boldsymbol{\beta} + c + u_1 \\ y_2 &= \mathbf{x}_2\boldsymbol{\beta} + c + u_2\end{aligned}$$

- Subtract $t = 1$ from $t = 2$ and define $\Delta y = y_2 - y_1$, $\Delta \mathbf{x} = \mathbf{x}_2 - \mathbf{x}_1$, and $\Delta u = u_2 - u_1$:

$$\Delta y = \Delta \mathbf{x} \boldsymbol{\beta} + \Delta u,$$

which is now a cross section in the changes or differences.

- We already know the conditions for OLS to be consistent for $\boldsymbol{\beta}$ when applied to this differenced equation: we just apply Assumptions OLS.1 and OLS.2:

$$\begin{aligned} E(\Delta \mathbf{x}' \Delta u) &= \mathbf{0} \\ \text{rank } E(\Delta \mathbf{x}' \Delta \mathbf{x}) &= K. \end{aligned}$$

- The rank condition is violated if $\mathbf{x}_t$ has elements that do not change over time because then the change is zero for all units.
- So elements of $\mathbf{x}_t$ that do not have time variation (that is, for at least some members in the population) get dropped out.
- The orthogonality condition $E(\Delta \mathbf{x}' \Delta u) = \mathbf{0}$ is

$$E[(\mathbf{x}_2 - \mathbf{x}_1)'(u_2 - u_1)] = \mathbf{0}.$$

But

$$\begin{aligned} E[(\mathbf{x}_2 - \mathbf{x}_1)'(u_2 - u_1)] &= E(\mathbf{x}_2' u_2) - E(\mathbf{x}_1' u_2) - E(\mathbf{x}_2' u_1) + E(\mathbf{x}_1' u_1) \\ &= -[E(\mathbf{x}_1' u_2) + E(\mathbf{x}_2' u_1)] \end{aligned}$$

because $E(\mathbf{x}_t' u_t) = \mathbf{0}$ if we maintain $E(u_t | \mathbf{x}_t, c) = 0$.

- Conclusion: Except by fluke, OLS on the differences will only be consistent if we add the assumptions

$$E(\mathbf{x}'_s u_t) = 0, s \neq t.$$

This is a kind of strict exogeneity assumption of the covariates with respect to the idiosyncratic errors.

- Assuming $\mathbf{x}_s$ is uncorrelated with u_t for all s and t is weaker than assuming $\mathbf{x}_s$ is uncorrelated with the composite error, $c + u_t$, for all s and t .

- Would we really omit an intercept from the differenced equation? Very unlikely.
- If we start with a model with different time intercepts,

$$\begin{aligned}y_1 &= \theta_1 + \mathbf{x}_1\boldsymbol{\beta} + c + u_1 \\y_2 &= \theta_2 + \mathbf{x}_2\boldsymbol{\beta} + c + u_2\end{aligned}$$

then

$$\Delta y = \alpha + \Delta \mathbf{x}\boldsymbol{\beta} + \Delta u,$$

where $\alpha = \theta_2 - \theta_1$ is *the change* in the aggregate time effects (intercepts).

- Testing $\alpha = 0$ might be of interest.
- Thus it is important to have a time dummy for each period (otherwise no intercept in the differenced equation)
- Now the rank condition also excludes variables (such as age) that change *by the same amount* for each unit.

Assumptions

- As mentioned earlier, we assume a balanced panel and all asymptotic analysis – implicit or explicit – is with fixed T and $N \rightarrow \infty$, where N is the size of the cross section.
- The basic *unobserved effects model* is

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}, \quad t = 1, \dots, T,$$

where $\mathbf{x}_{it}$ is $1 \times K$ and so $\boldsymbol{\beta}$ is $K \times 1$. In addition to unobserved effect and unobserved heterogeneity, c_i is sometimes called a *latent effect* or an individual effect, firm effect, school effect, and so on.

- An extension of the basic model is

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + \eta_t c_i + u_{it}, \quad t = 1, \dots, T,$$

where $\{\eta_t : t = 1, \dots, T\}$ are unknown parameters (and we have to assume something like $\eta_1 = 1$). More on this later.

- As in the earlier treatment, the model is written with $\boldsymbol{\beta}$ not depending on time. But $\mathbf{x}_{it}$ can include time period dummies and interactions of variables with time periods dummies, so the model is quite flexible.

- A general specification is

$$y_{it} = \mathbf{g}_t\boldsymbol{\theta} + \mathbf{z}_i\boldsymbol{\delta} + \mathbf{w}_{it}\boldsymbol{\gamma} + c_i + u_{it}$$

where $\mathbf{g}_t$ is a vector of aggregate time effects (often time dummies), $\mathbf{z}_i$ is a set of time-constant observed variables, and $\mathbf{w}_{it}$ changes across i and t (for at least some units i and time periods t). $\mathbf{w}_{it}$ can include interactions among time-constant and time varying variables.

- In microeconomic applications, best to avoid calling c_i a “random effect” or a “fixed effect.” We are treating c_i always as a random variable.

Assumptions about the Covariates and the Unobserved Effect

- In modern applications, “random effect” essentially means

$$\text{Cov}(\mathbf{x}_{it}, c_i) = \mathbf{0}, t = 1, \dots, T,$$

although we often will strengthen this.

- The term “fixed effect” means that no restrictions are placed on the relationship between c_i and $\{\mathbf{x}_{it}\}$.
- Recently, “correlated random effects” is used to denote situations where we model the relationship between c_i and $\{\mathbf{x}_{it}\}$, and it is especially useful for nonlinear models (but also for linear models, as we will see).

Assumptions about the Covariates and the Idiosyncratic Errors

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}$$

Contemporaneous Exogeneity Conditional on the Unobserved Effect:

$$E(u_{it} | \mathbf{x}_{it}, c_i) = 0$$

or

$$E(y_{it} | \mathbf{x}_{it}, c_i) = \mathbf{x}_{it}\boldsymbol{\beta} + c_i.$$

- Ideally, we could proceed with just this assumption.

- *Strict Exogeneity Conditional on the Unobserved Effect:*

$$E(y_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}, c_i) = E(y_{it} | \mathbf{x}_{it}, c_i) = \mathbf{x}_{it}\boldsymbol{\beta} + c_i$$

or $E(u_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}, c_i) = E(u_{it} | \mathbf{x}_{it}, c_i) = 0$

so that only $\mathbf{x}_{it}$ affects the expected value of y_{it} once c_i is controlled for.

- It is weaker than if we did not condition on c_i :

$$E(y_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}) = E(y_{it} | \mathbf{x}_{it}) = \mathbf{x}_{it}\boldsymbol{\beta},$$

- Assuming the condition holds conditional on c_i ,

$$E(y_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}) = \mathbf{x}_{it}\boldsymbol{\beta} + E(c_i | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}).$$

So correlation between c_i and $(\mathbf{x}_{i1}, \dots, \mathbf{x}_{iT})$ would invalidate the assumption without conditioning on c_i .

- But it is stronger than $E(u_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}) = 0$

- But strict exogeneity conditional on c_i rules out lagged dependent variables and feedback.
- Written in terms of the idiosyncratic errors, strict exogeneity is

$$E(u_{it} | \mathbf{x}_{i1}, \dots, \mathbf{x}_{iT}, c_i) = 0,$$

and so $\mathbf{x}_{i,t+h}$ must be uncorrelated with u_{it} for all $h > 0$.

- In addition to ruling out feedback, strict exogeneity assumes we have any distributed lag dynamics correct, too. For example, if $\mathbf{x}_{it} = (\mathbf{z}_{it}, \mathbf{z}_{i,t-1})$, then

$$E(y_{it} | \mathbf{z}_{i1}, \dots, \mathbf{z}_{it}, \dots, \mathbf{z}_{iT}, c_i) = E(y_{it} | \mathbf{z}_{it}, \mathbf{z}_{i,t-1}, c_i).$$

- A more reasonable assumption that we will use later is

$$E(y_{it} | \mathbf{x}_{it}, \mathbf{x}_{i,t-1}, \dots, \mathbf{x}_{i1}, c_i) = E(y_{it} | \mathbf{x}_{it}, c_i) = \mathbf{x}_{it}\boldsymbol{\beta} + c_i,$$

which is *sequential exogeneity conditional on the unobserved effect*.

- Sequential exogeneity assumes correct distributed lag dynamics but is silent on feedback.

Estimation and Testing

- The four most common estimation methods for unobserved effects models are pooled OLS, random effects, fixed effects, and first differencing.
- We have already covered POLS, and random effects is a particular GLS procedure.

Pooled OLS

- In discussing the properties of POLS, we just recognize that the equation is

$$\begin{aligned}y_{it} &= \mathbf{x}_{it}\boldsymbol{\beta} + v_{it} \\ v_{it} &= c_i + u_{it}\end{aligned}$$

- Consistency (fixed T , $N \rightarrow \infty$) of the POLS estimator is ensured by

$$E(\mathbf{x}'_{it} v_{it}) = \mathbf{0},$$

for which we should assume

$$\begin{aligned}E(\mathbf{x}'_{it} c_i) &= \mathbf{0} \\ E(\mathbf{x}'_{it} u_{it}) &= \mathbf{0}, t = 1, \dots, T.\end{aligned}$$

- Contemporaneous exogeneity $E(\mathbf{x}'_{it}u_{it}) = \mathbf{0}$ is weaker than strict exogeneity, but it buys us little in practice because POLS also uses $E(\mathbf{x}'_{it}c_i) = \mathbf{0}$, which cannot hold for lagged dependent variables and is unlikely for other variables that are not strictly exogenous.
- Inference should be made robust to serial correlation and heteroskedasticity.

- Let $\hat{v}_{it} = y_{it} - \mathbf{x}_{it}'\hat{\beta}_{POLS}$ be the POLS residuals. Then

$$\widehat{Avar}(\hat{\beta}_{POLS}) = \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}_{it}' \mathbf{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \sum_{r=1}^T \hat{v}_{it} \hat{v}_{ir} \mathbf{x}_{it}' \mathbf{x}_{ir} \right) \cdot \left(\sum_{i=1}^N \sum_{t=1}^T \mathbf{x}_{it}' \mathbf{x}_{it} \right)^{-1},$$

or sometimes with an adjustment, such as multiply by $N/(N-1)$.

- Can also write this estimator as

$$\widehat{Avar}(\hat{\beta}_{POLS}) = \left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i' \hat{v}_i \hat{v}_i' \mathbf{x}_i \right) \left(\sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1}$$

Random Effects Estimation

- State assumptions in conditional mean terms so that second moment derivations are easier.

ASSUMPTION RE.1:

$$\begin{aligned} \text{(a)} \quad E(u_{it} | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}, c_i) &= 0, \quad t = 1, \dots, T \\ \text{(b)} \quad E(c_i | \mathbf{x}_{i1}, \mathbf{x}_{i2}, \dots, \mathbf{x}_{iT}) &= E(c_i) \end{aligned}$$

- Assume $\mathbf{x}_{it}$ includes (at least) unity, and probably time dummies in addition. Then $E(c_i) = 0$ is without loss of generality.

- A GLS approach also leaves c_i in the error term:

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + v_{it}, \quad t = 1, 2, \dots, T$$

and we know the properties of feasible GLS when

$$E(\mathbf{x}'_{is} v_{it}) = 0, \quad \text{all } s, t = 1, \dots, T.$$

- This weaker version of strict exogeneity is implied by Assumption RE.1.

- Write the equation in system form (for all time periods) as

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + \mathbf{v}_i = \mathbf{X}_i\boldsymbol{\beta} + c_i\mathbf{j}_T + \mathbf{u}_i$$

where $\mathbf{j}'_T = (1, 1, \dots, 1)$.

- Define

$$\Omega_{T \times T} = E(\mathbf{v}_i\mathbf{v}'_i) = \text{Var}(\mathbf{v}_i).$$

ASSUMPTION RE.2: Ω is nonsingular and $\text{rank } E(\mathbf{X}'_i\Omega^{-1}\mathbf{X}_i)$.

- RE imposes a special structure on Ω (which could be wrong!).
- Under RE.1(a), c_i and u_{it} are uncorrelated.
- Assume further that

$$\begin{aligned} \text{Var}(u_{it}) &= \sigma_u^2, \quad t = 1, \dots, T \\ \text{Cov}(u_{it}, u_{is}) &= 0, \quad t \neq s \end{aligned}$$

Then

$$\begin{aligned} \text{Var}(v_{it}) &= \text{Var}(c_i + u_{it}) = \text{Var}(c_i) + \text{Var}(u_{it}) \\ \sigma_v^2 &= \sigma_c^2 + \sigma_u^2 \end{aligned}$$

- Further, for $t \neq s$,

$$\begin{aligned}
 \text{Cov}(v_{it}, v_{is}) &= \text{Cov}(c_i + u_{it}, c_i + u_{is}) \\
 &= \text{Var}(c_i) + \text{Cov}(c_i, u_{is}) + \text{Cov}(u_{it}, c_i) + \text{Cov}(u_{it}, u_{is}) \\
 &= \sigma_c^2
 \end{aligned}$$

- This leads to the “random effects” or “exchangeable” structure for Ω :

$$\begin{aligned}
 \Omega &= E[(c_i \mathbf{j}_T + \mathbf{u}_i)(c_i \mathbf{j}_T + \mathbf{u}_i)'] = E(c_i^2) \mathbf{j}_T \mathbf{j}_T' + E(\mathbf{u}_i \mathbf{u}_i') \\
 &= \sigma_c^2 \mathbf{j}_T \mathbf{j}_T' + \sigma_u^2 \mathbf{I}_T
 \end{aligned}$$

- Written in terms of its elements,

$$\Omega = \begin{pmatrix} \sigma_c^2 + \sigma_u^2 & \sigma_c^2 & \sigma_c^2 & \dots & \sigma_c^2 \\ \sigma_c^2 & \sigma_c^2 + \sigma_u^2 & \sigma_c^2 & \dots & \vdots \\ \vdots & \ddots & \ddots & \vdots & \sigma_c^2 \\ \sigma_c^2 & \dots & \sigma_c^2 & \sigma_c^2 + \sigma_u^2 & \sigma_c^2 \\ \sigma_c^2 & \dots & \sigma_c^2 & \sigma_c^2 & \sigma_c^2 + \sigma_u^2 \end{pmatrix},$$

so the $T \times T$ matrix depends on only two parameters, σ_c^2 and σ_u^2 or, more directly, $\sigma_v^2 = \sigma_c^2 + \sigma_u^2$ and σ_c^2 .

- Feasible GLS requires estimating Ω , that is, the two parameters.

- The pairwise correlations are $\text{Corr}(v_{it}, v_{is}) = \rho = \sigma_c^2 / (\sigma_c^2 + \sigma_u^2)$. Note that ρ is the fraction of the total variance accounted for by c_i . We can also write Ω as

$$\Omega = \sigma_v^2 \begin{pmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \rho \\ \rho & \cdots & \rho & 1 \end{pmatrix}$$

which shows we only need to estimate ρ to proceed with FGLS.

- Typically, we estimate σ_v^2 and σ_c^2 , but ρ is useful for summarizing the importance of c_i .

- We can use pooled OLS to get the residuals, $\check{v}_{it}$, across all i and t . Then a consistent estimator of σ_v^2 (not generally unbiased), as N gets large for fixed T , is

$$\hat{\sigma}_v^2 = (NT - K)^{-1} \sum_{i=1}^N \sum_{t=1}^T \check{v}_{it}^2 = SSR / (NT - K),$$

the usual variance estimator from OLS regression.

- This is based on, for each i , $\sigma_v^2 = T^{-1} \sum_{t=1}^T E(v_{it}^2)$ and then average across i , too.

- How to estimate σ_c^2 ? First, note that

$$\sigma_c^2 = [T(T-1)/2]^{-1} \sum_{t=1}^{T-1} \sum_{s=t+1}^T E(v_{it}v_{is}).$$

So a consistent “estimator” would be

$$\tilde{\sigma}_c^2 = [NT(T-1)/2]^{-1} \sum_{i=1}^N \sum_{t=1}^{T-1} \sum_{s=t+1}^T v_{it}v_{is}.$$

- An actual estimator replaces v_{it} with the POLS residuals,

$$\hat{\sigma}_c^2 = [NT(T-1)/2 - K]^{-1} \sum_{i=1}^N \sum_{t=1}^{T-1} \sum_{s=t+1}^T \check{v}_{it} \check{v}_{is},$$

which subtracts K from $NT(T-1)/2$ as a df adjustment. By the usual argument,

$$\text{plim}_{N \rightarrow \infty} \hat{\sigma}_c^2 = \sigma_c^2$$

with T fixed.

- Then $\hat{\sigma}_u^2 = \hat{\sigma}_v^2 - \hat{\sigma}_c^2$.

- Now we can use

$$\hat{\Omega} = \begin{pmatrix} \hat{\sigma}_v^2 & \dots & \hat{\sigma}_c^2 & \hat{\sigma}_c^2 \\ \hat{\sigma}_c^2 & \hat{\sigma}_v^2 & & \hat{\sigma}_c^2 \\ \vdots & & \ddots & \vdots \\ \hat{\sigma}_c^2 & \dots & \hat{\sigma}_c^2 & \hat{\sigma}_v^2 \end{pmatrix} \text{ or } \hat{\Lambda} = \begin{pmatrix} 1 & \dots & \hat{\rho} & \hat{\rho} \\ \hat{\rho} & 1 & & \hat{\rho} \\ \vdots & & \ddots & \vdots \\ \hat{\rho} & \dots & \hat{\rho} & 1 \end{pmatrix},$$

where $\hat{\rho} = \hat{\sigma}_c^2 / \hat{\sigma}_v^2$, in FGLS.

- It is possible for $\hat{\sigma}_c^2$ to be negative, which means the basic unobserved effects variance-covariance structure is faulty.
- Typically, $\hat{\sigma}_c^2 > 0$ unless the variables have been transformed in some way – such as being first differenced – before applying GLS.
- The FGLS estimator that uses this particular structure of $\hat{\Omega}$ is the *random effects (RE) estimator*.

- Fully robust inference is available for RE, and there are good reasons for doing so.

(1) Ω may not have the special (and restrictive, especially for large T) RE structure, that is, $E(\mathbf{v}_i \mathbf{v}_i')$ need not have the RE form. Serial correlation or changing variances in $\{u_{it} : t = 1, \dots, T\}$ invalidate the RE structure.

(2) The system homoskedasticity requirement,

$$E(\mathbf{v}_i \mathbf{v}_i' | \mathbf{X}_i) = E(\mathbf{v}_i \mathbf{v}_i')$$

might not hold.

- A fully robust estimator is

$$\widehat{Avar}(\hat{\beta}_{RE}) = \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{v}_i \mathbf{v}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right) \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1},$$

where $\mathbf{v}_i = \mathbf{y}_i - \mathbf{X}_i \hat{\beta}_{RE}$ is the vector of RE (FGLS) residuals.

- Sometimes, an iterative procedure is used. New residuals can be used to obtain a new estimate of Ω , and so on.
- Using first-order asymptotics, no efficiency gain from iterating. Might help with smaller N , though.

- If we are allowing for the fact that $E(\mathbf{v}_i\mathbf{v}_i')$ does not have the RE form, or that $E(\mathbf{v}_i\mathbf{v}_i'|\mathbf{X}_i)$ is not constant, why use RE instead of pooled OLS?
- Fully robust asymptotic variance estimators are available for both, so the choice is not based on valid inference.
- The key is potential efficiency gains using RE even though we may not be doing true (feasible) GLS. RE accounts for some form of serial correlation in estimation. POLS ignores serial correlation entirely.

- In comparing the standard errors for RE and POLS it only makes sense to compare fully robust standard errors.
- The reason for using RE rather than POLS is the same motivation as used in the generalized estimating equations literature: some inefficient procedures may nevertheless be more efficient than other inefficient procedures.

- What is the advantage of RE, which imposes specific assumptions on Ω , over the unrestricted FGLS we discussed earlier?
- Theoretically, nothing. We do not get more efficiency with large N and small T by imposing restrictions on Ω .
- If system homoskedasticity holds but Ω is not of the RE form, an unrestricted FGLS analysis is more efficient than RE (again, fixed T , $N \rightarrow \infty$).
- As we will see later, RE does have some appeal because of its implicit transformation.

- A nonrobust variance matrix estimator can be used if we add second moment assumptions:

ASSUMPTION RE.3:

$$(a) E(\mathbf{u}_i \mathbf{u}_i' | \mathbf{x}_i, c_i) = \sigma_u^2 \mathbf{I}_T$$

$$(b) E(c_i^2 | \mathbf{x}_i) = \sigma_c^2$$

- Under Assumptions RE.1 and RE.3, Ω has the RE structure *and* system homoskedasticity holds.
- Part (a) is homoskedasticity and serial uncorrelatedness of $\{u_{it}\}$ conditional on $(\mathbf{x}_i, c_i)$, and (b) is homoskedasticity of c_i .

- Under RE.1, RE.2, and RE.3,

$$\widehat{Avar}(\hat{\beta}_{RE}) = \left(\sum_{i=1}^N \mathbf{x}_i' \hat{\Omega}^{-1} \mathbf{x}_i \right)^{-1}$$

is a valid asymptotic variance matrix estimator.

- Inference is straightforward. Typically use Wald or robust Wald statistic for multiple restrictions.

Fixed Effects Estimation

- Unlike POLS and RE, fixed effects estimation removes c_i to form an estimating equation.
- Average the original equation,

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}, \quad t = 1, \dots, T,$$

across t to get a cross-sectional equation:

$$\bar{y}_i = \bar{\mathbf{x}}_i\boldsymbol{\beta} + c_i + \bar{u}_i,$$

where the overbar indicates *time averages*:

$$\bar{y}_i = T^{-1} \sum_{t=1}^T y_{it}, \quad \bar{\mathbf{x}}_i = T^{-1} \sum_{t=1}^T \mathbf{x}_{it}, \quad \bar{u}_i = T^{-1} \sum_{t=1}^T u_{it}$$

- The equation $\bar{y}_i = \bar{\mathbf{x}}_i\boldsymbol{\beta} + c_i + \bar{u}_i$ is often called the *between equation* because it relies on variation in the data between cross section observations.
- The *between estimator* is the OLS estimator from the cross section regression

$$\bar{y}_i \text{ on } \bar{\mathbf{x}}_i, i = 1, \dots, N.$$

[In practice, an intercept is included to account for nonzero $E(c_i)$.]

- The between estimator is inconsistent unless

$$\text{Cov}(\bar{\mathbf{x}}_i, c_i) = \mathbf{0}, \text{Cov}(\bar{\mathbf{x}}_i, \bar{u}_i) = \mathbf{0}.$$

- Instead, subtract off the time-averaged equation from the original equation to eliminate c_i :

$$y_{it} - \bar{y}_i = (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)\boldsymbol{\beta} + u_{it} - \bar{u}_i, \quad t = 1, \dots, T$$

or

$$\ddot{y}_{it} = \ddot{\mathbf{x}}_{it}\boldsymbol{\beta} + \ddot{u}_{it}, \quad t = 1, \dots, T$$

where $\ddot{y}_{it} = y_{it} - \bar{y}_i$ and so on.

- We call this the *time demeaned equation*, and the transformation is *time demeaning*, *fixed effects*, or *within* (time variation within each i is used).

- Key is that c_i is gone from the time demeaned equation.
- So, we can use pooled OLS:

$$\ddot{y}_{it} \text{ on } \ddot{x}_{it}, \quad t = 1, \dots, T; i = 1, \dots, N.$$

This is the *fixed effects (FE) estimator* or the *within estimator*.

$$\begin{aligned} \hat{\beta}_{FE} &= \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{y}_{it} \right) \\ &= \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' y_{it} \right) \end{aligned}$$

because $\sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' (y_{it} - \bar{y}_i) = \sum_{t=1}^T (\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' y_{it}$.

- What is the weakest orthogonality assumption for consistency? We can just apply the results for POLS, but it is useful to see it directly.
- Write the estimator by substituting $\ddot{y}_{it} = \ddot{x}_{it}\beta + \ddot{u}_{it}$:

$$\begin{aligned}\hat{\beta}_{FE} &= \beta + \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{u}_{it} \right) \\ &= \beta + \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' u_{it} \right)\end{aligned}$$

- By the WLLN as $N \rightarrow \infty$ with fixed T , the key moment condition for consistency is

$$\sum_{t=1}^T E(\ddot{\mathbf{x}}'_{it} u_{it}) = \sum_{t=1}^T E[(\mathbf{x}_{it} - \bar{\mathbf{x}}_i)' u_{it}] = \mathbf{0}.$$

- In addition to contemporaneous exogeneity, $E(\mathbf{x}'_{it} u_{it}) = \mathbf{0}$, we need a kind of strict exogeneity:

$$E(\bar{\mathbf{x}}'_i u_{it}) = T^{-1} \sum_{s=1}^T E(\mathbf{x}'_{is} u_{it}) = \mathbf{0}, \quad t = 1, 2, \dots, T.$$

ASSUMPTION FE.1: Same as RE.1(a), that is,

$$E(u_{it}|\mathbf{x}_i, c_i) = 0, \quad t = 1, \dots, T.$$

- This implies $E(\mathbf{x}'_{is}u_{it}) = \mathbf{0}$, all $s, t = 1, \dots, T$, and so $E(\ddot{\mathbf{x}}'_{it}u_{it}) = \mathbf{0}$, $t = 1, \dots, T$.
- The rank condition is directly from POLS.2:

ASSUMPTION FE.2:

$$\text{rank} \left[\sum_{t=1}^T E(\ddot{\mathbf{x}}'_{it}\ddot{\mathbf{x}}_{it}) \right] = K.$$

- The rank condition rules out elements in $\mathbf{x}_{it}$ that have no time variation for any unit in the population. Such variables get swept away by the within transformation.
- Under FE.1 and FE.2,

$$\hat{\beta}_{FE} \xrightarrow{p} \beta \text{ as } N \rightarrow \infty$$

- What parameters can we identify with FE?
- Suppose we start with

$$y_{it} = \theta_1 + \theta_2 d2_t + \dots + \theta_T dT_t + \mathbf{z}_i \gamma_1 + d2_t \mathbf{z}_i \gamma_2 + \dots + dT_t \mathbf{z}_i \gamma_T + \mathbf{w}_{it} \delta + c_i + u_{it}$$

- Using FE, we cannot estimate θ_1 or γ_1 , but all other parameters are generally identified.
- FE allows c_i to be arbitrarily correlated with $(\mathbf{z}_i, \mathbf{w}_{it})$, and so we cannot distinguish $\theta_1 + \mathbf{z}_i \gamma_1$ from c_i .
- We can estimate $\theta_2, \dots, \theta_T$ and $\gamma_2, \dots, \gamma_T$. So we can estimate whether the effect of the time constant variables has changed over time. We cannot estimate the effect in any period t because it is γ_1 for $t = 1$ and $\gamma_1 + \gamma_t$ for $t = 2, \dots, T$.

- We can obtain a variance matrix estimator valid under Assumptions FE.1 and FE.2.
- Define the FE residuals as

$$\hat{u}_{it} = \ddot{y}_{it} - \ddot{x}_{it}\hat{\beta}_{FE}, \quad t = 1, \dots, T; i = 1, \dots, N$$

- These are “estimates” of the $\ddot{u}_{it}$, not the u_{it} . This has implications for estimating the error variance, σ_u^2 .
- Without additional assumptions, use the “cluster-robust” matrix

$$\begin{aligned} \widehat{Avar}(\hat{\beta}_{FE}) &= \left(\sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \left(\sum_{i=1}^N \ddot{X}_i' \hat{u}_i \hat{u}_i' \ddot{X}_i \right) \left(\sum_{i=1}^N \ddot{X}_i' \ddot{X}_i \right)^{-1} \\ &= \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1} \left(\sum_{i=1}^N \sum_{t=1}^T \sum_{r=1}^T \hat{u}_{it} \hat{u}_{ir}' \ddot{x}_{it}' \ddot{x}_{ir} \right) \\ &\quad \cdot \left(\sum_{i=1}^N \sum_{t=1}^T \ddot{x}_{it}' \ddot{x}_{it} \right)^{-1}. \end{aligned}$$

- Of course, a nonrobust form requires an extra assumption:

ASSUMPTION FE.3: Same as RE.3(a), that is,

$$E(\mathbf{u}_i \mathbf{u}_i' | \mathbf{x}_i, c_i) = \sigma_u^2 \mathbf{I}_T.$$

- So $Avar[\sqrt{N}(\hat{\beta}_{FE} - \beta)] = \sigma_u^2 [E(\ddot{X}_i' \ddot{X}_i)]^{-1}$ under FE.1, FE.2, and FE.3.
- Estimating σ_u^2 requires some care because we effectively observe $\ddot{u}_{it}$, not u_{it} .

- The FE estimator $\hat{\beta}_{FE}$ can also be obtained by running a long regression on the original data, and including dummy variables for each cross section unit:

$$y_{it} \text{ on } d1_i, d2_i, \dots, dN_i, \mathbf{x}_{it}, \quad t = 1, \dots, T; i = 1, \dots, N,$$

often called the *dummy variable regression*.

- Only danger: Treating the c_i as parameters to estimate, while sensible with “large” T , can lead to trouble later with nonlinear models. Here, we get a consistent estimator of β for fixed T .
- Sometimes we want to estimate the c_i using the T time periods. Do not have to run the dummy variable regression:

$$\hat{c}_i = \bar{y}_i - \bar{\mathbf{x}}_i \hat{\beta}_{FE}, \quad i = 1, \dots, N.$$

- With small T , this is not a good “estimate” of c_i (the right hand side above involves averages across t)

Fixed Effects GLS

- Write the T time periods for a random draw i as

$$\mathbf{y}_i = \mathbf{X}_i\boldsymbol{\beta} + c_i\mathbf{j}_T + \mathbf{u}_i$$

and let the variance matrix of $\mathbf{u}_i$ to be a $T \times T$ unrestricted matrix Λ .

- When we eliminate c_i by demeaning we get

$$\mathbf{y}_i = \ddot{\mathbf{X}}_i\boldsymbol{\beta} + \ddot{\mathbf{u}}_i$$

where, for example, $\ddot{\mathbf{u}}_i = \mathbf{Q}_T\mathbf{u}_i$ and $\mathbf{Q}_T = \mathbf{I}_T - \mathbf{j}_T(\mathbf{j}'_T\mathbf{j}_T)^{-1}\mathbf{j}'_T$ is symmetric, idempotent with rank $T - 1$.

- Because $\mathbf{j}'_T \ddot{u}_i = 0$, we know the $T \times T$ matrix $E(\ddot{u}_i \ddot{u}'_i)$ has rank less than T . In fact, (unconditional) variance covariance matrix of $\ddot{u}_i$ is

$$\Omega \equiv E(\ddot{u}_i \ddot{u}'_i) = E(\mathbf{Q}_T \mathbf{u}_i \mathbf{u}'_i \mathbf{Q}_T) = \mathbf{Q}_T \Lambda \mathbf{Q}_T,$$

which has rank $T - 1$.

- Applying FGLS to

$$\mathbf{y}_i = \ddot{X}_i \boldsymbol{\beta} + \ddot{u}_i$$

is tricky.

- There is a simple solution.
- After demeaning to obtain $\mathbf{y}_i$ and $\ddot{X}_i$ using all T time periods and obtaining

$$\hat{\Omega} = N^{-1} \sum_{i=1}^N \hat{u}_i \hat{u}_i'$$

drop one of the time periods. It does not matter which one is dropped (but the first or last are easiest).

- Apply FGLS to the $T - 1$ remaining equations using $\hat{\Omega}$.
- Remember, can still make a case for robust inference because system heteroskedasticity is always a possibility.

Some Practical Hints in Applying Fixed Effects

- Suppose i is a firm. Then the phrase “I included firm fixed effects” means that c_i is allowed to be correlated with the covariates. If c_i is called a “firm random effect” then c_i is being assumed to be uncorrelated with $\mathbf{x}_{it}$.
- Suppose that we cannot, or do not want to, use FE estimation. This might occur because the key variable at the firm level is constant across time for all firms – and so the FE transformation sweeps it away – or there is little time variation within firm in the key variable, leading to large standard errors.
- Instead, we might use a random effects analysis at the firm level but include industry dummy variables to account for systematic differences across industries. So, we include in $\mathbf{x}_{it}$ a set of industry dummy variables while also allowing a firm effect c_i in a “random effects” analysis.
- If there are many firms per industry, the industry “fixed effects” – the coefficients on the industry dummies – can be precisely estimated. So the industry “fixed effects” are really parameters to estimate whereas the c_i are not.

- Generally, including dummies for more aggregated levels and then applying RE is common when the covariates of interest vary in the cross section but not (much) over time.
- Keep in mind that an RE analysis at the firm level with industry dummies need not be entirely convincing: the key elements of $\mathbf{x}_{it}$ might be correlated with unobserved firm features that are not adequately captured by industry differences.

First-Differencing Estimation

- Like FE, FD removes c_i .
- But it does it by differencing adjacent observations. FE and FD are the same when $T = 2$, but differ otherwise.
- Again, start with the original equation:

$$y_{it} = \mathbf{x}_{it}\boldsymbol{\beta} + c_i + u_{it}, t = 1, \dots, T.$$

For FD, we explicitly lose the first time period:

$$\Delta y_{it} = \Delta \mathbf{x}_{it}\boldsymbol{\beta} + \Delta u_{it}, t = 2, \dots, T.$$

The FD estimator is pooled OLS on the first differences.

- In practice, might not difference period dummies, unless interested in the year intercepts in the original levels.

- FD also requires a kind of strict exogeneity. The weakest assumption is

$$E(\Delta \mathbf{x}_{it}' \Delta u_{it}) = 0, \quad t = 2, \dots, T.$$

- (For later: In unbalanced cases, FD requires that data exists in adjacent time periods. FE does not.)
- A sufficient condition is

ASSUMPTION FD.1: Same as FE.1, $E(u_{it} | \mathbf{x}_i, c_i) = 0, \quad t = 1, \dots, T.$

ASSUMPTION FD.2: Let $\Delta \mathbf{X}_i$ be the $(T - 1) \times K$ matrix with rows $\Delta \mathbf{x}_{it}$. Then,

$$\text{rank } E(\Delta \mathbf{X}_i' \Delta \mathbf{X}_i) = K.$$

- Should make inference robust to serial correlation and heteroskedasticity in the differenced errors, $e_{it} \equiv u_{it} - u_{i,t-1}$. For example, if $\{u_{it}\}$ is uncorrelated, $\text{Corr}(e_{it}, e_{i,t+1}) = -.5$.
- After POLS on the first differences, let

$$\hat{e}_{it} = \Delta y_{it} - \Delta \mathbf{x}_{it}' \hat{\beta}_{FD}, \quad t = 2, \dots, T; i = 1, \dots, N$$

and let $\mathbf{\hat{e}}_i = (\hat{e}_{i2}, \dots, \hat{e}_{iT})'$ be the $(T-1) \times 1$ residuals. Then

$$\widehat{\text{Avar}}(\hat{\beta}_{FD}) = \left(\sum_{i=1}^N \Delta \mathbf{x}_i' \Delta \mathbf{x}_i \right)^{-1} \left(\sum_{i=1}^N \Delta \mathbf{x}_i' \mathbf{\hat{e}}_i \mathbf{\hat{e}}_i' \Delta \mathbf{x}_i \right) \left(\sum_{i=1}^N \Delta \mathbf{x}_i' \Delta \mathbf{x}_i \right)^{-1}$$

is the fully robust variance matrix estimator.

- Applying feasible GLS after differencing is especially easy because the lost degree of freedom for each i is automatically incorporated by losing the first time period.
- Resulting estimator is the FDGLS estimator. It uses an unrestricted $(T - 1) \times (T - 1)$ variance matrix in the FD equation

$$\Delta \mathbf{y}_i = \Delta \mathbf{X}_i \boldsymbol{\beta} + \Delta \mathbf{u}_i$$

where $\Delta \mathbf{u}_i$ is $(T - 1) \times 1$.